Pembrolizumab, capecitabine and oxaliplatin

Pembrolizumab, capecitabine and oxaliplatin is a cancer drug treatment. It is used to treat cancer of the oesophagus or gastro-oesophageal junction.

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What is pembrolizumab, capecitabine and oxaliplatin?

Pembrolizumab, capecitabine and oxaliplatin is a combination of cancer drugs. It is also called pembrolizumab with CAPOX, or pembrolizumab with XELOX.

It is used to treat cancer of the oesophagus or gastro-oesophageal junction.

- pembrolizumab is a type of immunotherapy drug called a checkpoint inhibitor
- capecitabine is a chemotherapy drug
- oxaliplatin is a chemotherapy drug

It is best to read this information with our general information about cancer drug treatments and the type of cancer you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree (consent) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

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DPD testing before treatment

Before starting this treatment, you should have a blood test to check for gene changes that affect how your body produces DPD. DPD is an enzyme that helps the body break down capecitabine.

If you have low DPD levels or no DPD (DPD deficiency), you have more risk of serious or life-threatening side effects from capecitabine. To reduce your risk, your doctor may give you:

- a lower dose of capecitabine to start with
- a different type of chemotherapy.

Having low DPD levels does not usually cause any symptoms. Without a test, you will not know whether your DPD levels are low. Testing finds most people who are affected, but not all. Your doctor, nurse or pharmacist can give you more information about your risk of DPD deficiency, before you start treatment.

Having pembrolizumab, capecitabine and oxaliplatin

You have pembrolizumab and oxaliplatin as drips (infusions) into a vein (intravenously). Or you may have pembrolizumab as an injection under the skin (subcutaneously). You usually have pembrolizumab and oxaliplatin in a chemotherapy day unit as an outpatient. Capecitabine comes in tablets. This means you can take it at home.

During a course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your treatment.

Your nurse will usually give you anti-sickness drugs before the cancer drugs. Your nurse may give pembrolizumab as an injection into your thigh or tummy (abdomen) over several minutes. Or they will give both oxaliplatin and pembrolizumab through one of the following:

- a cannula – a short, thin tube the nurse puts into a vein in the arm or hand
- a central line – a fine tube that goes under the skin of the chest and into a vein close by
- a PICC line – a fine tube that is put into a vein in the arm and goes up into a vein in the chest
- an implantable port (portacath) – a disc that is put under the skin on the chest or arm and goes into a vein in the chest.

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Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take these exactly as they tell you to.

Your course of treatment

You usually have a course of several cycles of treatment over a few months. Your doctor, nurse or pharmacist will discuss your treatment plan with you.

This treatment may be given for up to 2 years. You have all 3 drugs to start with. You have:

- pembrolizumab once every 3 or 6 weeks
- oxaliplatin once every 3 weeks
- capecitabine tablets twice a day. Your cancer team will explain which days you should take the tablets.

Oxaliplatin and capecitabine are only given for up to 24 weeks. After this you continue to have pembrolizumab once every 3 or 6 weeks for up to 2 years.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take these exactly as they tell you to.

Taking capecitabine tablets

Your cancer team will give you the capecitabine tablets to take home. They may give you tablets of different strengths. Always take the tablets exactly as explained. This is important to make sure they work as well as possible for you. Depending on the side effects, they may change your dose of capecitabine. It is very important that you know how many tablets to take if the dose is changed.

Make sure you:

- take your tablets twice a day, about 10 to 12 hours apart within 30 minutes after a meal. Take the tablets in the morning after breakfast and then after your evening meal. Try to take them at the same times every day.
- swallow them whole with a glass of water.
- do not chew, break or crush them

Ask your doctor, nurse or pharmacist for advice if you find it difficult to swallow the tablets, or if you use a feeding tube. They can give you instructions about dissolving the tablets. It is important to follow their instructions carefully.

If you forget to take the tablets, skip the forgotten dose and take the next dose at the usual time. Do not take a double dose to make up for the missed dose. You can contact the hospital on the 24-hour number you have been given for advice.

Other things to remember about your tablets:

- Wash your hands after taking them.
- Other people should avoid direct contact with them.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.

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- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- Do not throw away unused capsules or tablets. Return them to your cancer team at the hospital.

Lactose

Capecitabine tablets contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common. This treatment can make the immune system too active. This can cause side effects that may be serious if not treated quickly.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Do not try to treat side effects yourself. Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Delayed side effects of immunotherapy drugs

Side effects can start during treatment but some may start weeks, months, or sometimes more than a year after you finish treatment.

If you have symptoms or side effects that start during treatment or after treatment ends, contact the hospital straight away on the 24-hour number. Side effects can become serious if they are not treated.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

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Your patient alert card

They will also give you a card with information about your treatment. It is important to keep the card with you during treatment and after treatment ends. You should show it to any doctor or healthcare professional caring for you.

Side effects while treatment is being given

Some people have side effects while they are having a cancer drug or shortly after they have it.

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Throat spasm

Rarely, oxaliplatin can cause a spasm in the throat area around the voicebox (larynx). This can cause difficulties with swallowing and breathing. This might happen during treatment, or in the first few days after treatment.

This side effect can be frightening, but it should only be temporary. If you have breathing difficulties, take long, deep breaths through your nose. This will have a calming effect and help your breathing return to normal.

This symptom may be worse in cold temperatures. During treatment, and for a few days afterwards, it may help to:

- avoid cold drinks and ice cubes
- wrap up warm and cover your nose and mouth in cold weather.

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It is important to tell your doctor, nurse or pharmacist if you have this side effect. They may give the next doses of oxaliplatin over 4 to 6 hours to reduce the risk of throat spasms.

The drug leaks outside the vein

Sometimes cancer drugs that are given into a vein may leak outside the vein. If this happens, some drugs can damage the tissue near the vein. This is called extravasation. Extravasation is not common, but it is important that it is dealt with quickly. If you have any of the following symptoms around the vein during or after your treatment, tell your nurse straight away:

- stinging
- pain
- swelling
- if you have white skin, the area may become red
- if you have black or brown skin, the area may become darker.

Injection site

If you have pembrolizumab as a subcutaneous injection, the skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin

If this happens, let your doctor know. Painkillers may help.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this combined treatment.

We also have information about the individual side effects of:

- chemotherapy drugs capecitabine and oxaliplatin
- immunotherapy drug pembrolizumab

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an infection. A low white blood cell count is called neutropenia.

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An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
-

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breathless

- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Diarrhoea

Capecitabine and pembrolizumab may cause diarrhoea. Sometimes it is severe. Diarrhoea means passing more stools (poo) than is normal for you or having watery or loose stools. If you have a stoma, it will be more active than usual. Your doctor, nurse or pharmacist may give you anti-diarrhoea drugs to take at home.

If you have diarrhoea or an increase in stoma activity, follow their advice on:

- taking the anti-diarrhoea drugs
- how much and what type of fluids to drink
- any changes to your diet that might help.

After this, contact the hospital on the 24-hour number straight away if any of the following things happen:

- you have diarrhoea at night
- you have uncomfortable stomach cramps
- you have diarrhoea 4 or more times in a day
- you have a moderate increase in stoma activity
- the anti-diarrhoea drugs do not work within 24 hours.

They will tell you to stop taking capecitabine. When the diarrhoea is better, they will tell you whether you can start taking it again. Sometimes they reduce the dose.

This treatment can also cause inflammation of the bowel (colitis) which can be very serious.

Contact the hospital straight away on the 24-hour number if you have any symptoms above or any of the following symptoms during treatment, or after treatment ends:

- You have uncomfortable tummy cramps.
- Your temperature is over 37.5°C.
- There is blood or mucus in your stool.

You may need to go into hospital to have fluids through a drip or treatments such as steroids.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

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If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called peripheral neuropathy. You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

During treatment, these symptoms are often made worse by the cold. It can help to keep warm and avoid iced drinks. You may need to take care opening the fridge and use gloves to handle frozen food.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about coping with changes to taste.

Sore mouth and throat

This treatment may cause a sore mouth and throat. You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

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Loss of appetite

This treatment can affect your appetite. Don’t worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Constipation

This treatment can cause constipation. Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- Eat high-fibre foods, such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Hair loss

Your hair may get thinner. But you are unlikely to lose all the hair from your head. Hair loss usually starts after your first or second treatment. It is almost always temporary, and your hair will usually grow back after treatment ends.

Your nurse can talk to you about ways to cope with hair loss.

Skin changes

This treatment may cause inflammation in the skin. This is usually mild, but if it is not treated it can become serious. You may need creams, steroids or other medicines to treat skin problems.

Contact the hospital on the 24-hour number as soon as possible, if you notice any skin changes during treatment, or after treatment ends. This includes:

- a rash or bumps on the skin
- dry, tight or itchy skin
- areas that are a different colour to your usual skin tone.

Here are some things you can do to take care of your skin during treatment:

- Wash with lukewarm water and mild, unperfumed, soap-free cleansers.
- Moisturise your skin regularly with unperfumed moisturisers for dry skin conditions.
- Use sun cream of at least SPF 30 to protect your skin from the sun. Cover up with clothing and a hat.

Contact the hospital straight away on the 24-hour number if you have any of these serious skin symptoms:

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- a rash that is spreading
- swollen, blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain.

Muscle and joint pain

This treatment can cause sore or weak muscles. Your joints may also be stiff, sore or swollen. This is called arthritis. If you already have problems with joint pain, this may become worse.

Contact the hospital on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- new or worsening aches or pains
- new or worsening stiffness in a joint
- swollen joints or muscle weakness.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Feeling tired

Feeling tired is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Effects on sleep or mood

Sometimes this treatment can affect your sleep or your mood. Tell your doctor, nurse or pharmacist if you notice this or if cancer treatment makes it difficult to sleep.

Hormone changes

This treatment may permanently affect glands in the body that produce hormones. Hormones control different processes in the body. You will have regular blood tests to check some hormone levels, such as your thyroid. You may need to take tablets daily to correct your hormone levels.

Contact the hospital on the 24-hour number if you have any of these side effects during treatment or after it ends:

- increased sweating
- problems with constipation
- weight gain or loss
- dizziness or fainting
- feeling or being sick

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- feeling more hungry or thirsty than usual
- loss of sex drive
- passing urine (peeing) more often than usual
- headaches that do not go away
- feeling more tired than usual (fatigue)
- mood changes
- changes to your heartbeat (palpitations).

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Effects on the lungs

This treatment can cause inflammation of the lungs. This is called pneumonitis. Contact the hospital straight away on the 24-hour number if you notice any of these changes during treatment or after it ends:

- breathlessness
- a cough that does not go away
- wheezing
- a fever, with a temperature over 37.5°C.

You should also tell them if any existing breathing problems get worse. You may have:

- tests to check your lungs
- treatment, if needed.

Hand-foot (palmar-plantar) syndrome

This treatment can affect the palms of your hands and the soles of your feet. This is called palmar-plantar or hand-foot syndrome.

If you have white skin these areas may become red. If you have black or brown skin, these areas might get darker.

The skin on the palms of your hands and the soles of your feet may:

- be sore
- be painful, tingle, or swell
- peel, crack or blister.

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If you have any of these symptoms, contact the hospital straight away on the 24-hour number. They can give you advice. This is especially important if you have any broken skin or if walking is difficult. They can prescribe creams and painkillers to help.

You can care for your hands and feet by:

- gently moisturising your hands and feet regularly
- wearing gloves to protect your hands and nails when working in the house or garden
- wearing loose cotton socks and avoiding tight-fitting shoes and gloves.

Nail changes

This treatment can affect your nails. They may grow more slowly or break more easily. You might notice ridges or white or dark lines across your nails. These changes usually disappear as the nails grow out after treatment. Sometimes nails can become loose or fall off. Tell your doctor, nurse or pharmacist about any changes to your nails.

If the skin around your nails becomes sore and swollen, contact the hospital straight away on the 24-hour number. These might be signs of an infection.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Effects on the heart

Pembrolizumab and capecitabine can both affect how the heart works. This is not common, but it can be serious. You may have tests to check how well your heart is working. These may be done before, during and after treatment.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat
- swollen feet and ankles.

If you are still taking capecitabine and you have any of these symptoms, stop taking the capecitabine tablets. Do not start taking it again until your doctor, nurse or pharmacist tells you to.

Always call 999 if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

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Effects on the liver

This treatment may affect how your liver works. It can cause liver inflammation called hepatitis. You will have regular blood tests to check how your liver is working. Tell your doctor, nurse or pharmacist if you have loss of appetite.

Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Bladder effects

Oxaliplatin may cause you discomfort when you pass urine. You may also pass urine more often than usual. Make sure you drink plenty of fluids during the 24 hours following chemotherapy. Try to drink at least 2 litres (3½ pints).

It is also important to empty your bladder regularly and to try to pass urine as soon as you feel the need to go.

Contact the hospital straight away if you feel any discomfort or stinging when you pass urine, or if you notice any blood in it.

Hearing changes

This treatment may cause hearing changes, including hearing loss. You may have ringing in the ears. This is called tinnitus. You may also become unable to hear some high-pitched sounds. Hearing changes may get better after this treatment ends. But this does not always happen. If you notice any changes in your hearing, tell your doctor, nurse or pharmacist.

Effects on the kidneys

This treatment can affect how the kidneys work. It may cause kidney inflammation called nephritis. You will have regular blood tests to check how your kidneys are working.

Contact the hospital straight away on the 24-hour number during treatment, or after it ends, if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

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Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Eyesight changes

This treatment can affect your eyes and eyesight. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- dry, itchy, painful, or swollen eyes
- eyes that are sensitive to light
- changes to your eyesight.

Effects on the nervous system

Rarely, this treatment can affect the brain and nerves. These effects may be mild at first but can become serious if not treated. Symptoms may include:

- weakness, numbness or tingling in your arms, legs or face
- a headache that does not get better
- seeing double or having drooping eyelids
- difficulty turning or bending your nec
- drowsiness or confusion
- problems with moving or speaking
- fits (seizures).

If you have any of these symptoms during treatment or after it ends, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away
- call **999**.

Do not drive yourself to hospital.

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Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a blood clot. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and coronavirus (covid). They may also recommend other vaccines, such as Shingrix® for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

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Contraception

Your doctor, nurse or pharmacist will advise you not to get pregnant or make someone pregnant while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex during this treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Show them your patient alert card. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist or optician, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

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Author

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

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Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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